

# Stephen Purdue

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## EDUCATION

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### Royal Holloway, University of London

*MSc, Artificial Intelligence*

London, UK

2026 - Expected 2027

### University of Winchester

*BSc, Computer Game Development*

Winchester, UK

2023 - 2026

## EXPERIENCE

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### Founder

*Corvian, London, UK*

2026 - Present

- Drove 100% of technical strategy and execution, leading the design and creation of a scalable kitchen management system using Python, FastAPI, and TypeScript.
- Implemented critical systems including containerisation via Docker, AWS Caching and Load Balancing, and Stripe payments.

### Sous-Chef, Chef de Partie

*Mustang Group, Southampton, UK*

2021 - Present

- Led a kitchen of five through high-volume, deadline-driven service under constant time pressure, requiring rapid problem-solving, attention to detail, and direct communication with other colleagues in an agile environment.
- Mentored and trained 4-8 junior chefs and new starters per intake, maintaining consistent service standards and reducing onboarding time during high staff turnover.

## PROJECTS

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### Stock Market Prediction Engine

[GitHub](#)

- Developed an LSTM model in PyTorch to forecast daily stock prices from historical time-series analysis data, reducing prediction error by 81% versus a moving-average baseline. Containerised the full stack with Docker Compose, reducing local setup time to under 5 minutes.

### UNIX Shell

[GitHub](#)

- Developed a UNIX style shell in Python, implementing command parsing, built-in commands, environment-variable handling, and input/output redirection. Developed a modular, testable architecture using OS-level process and file-system APIs, strengthening my understanding of Linux systems programming.

### Procedural Generation Algorithm

[GitHub](#)

- Engineered a scalable procedural generation system in Unity using C# and Binary Space Partitioning to recursively subdivide space into room layouts and connective corridors in linear time. Reduced level build-time by 77% by implementing modular, parameter-driven logic.

### Adaptive Combat Simulator

[GitHub](#)

- Developed an adaptive combat simulator, achieving a 75% win rate against a baseline agent, with overall reward improving from 32% to 87%, measured over 30 unique evaluation simulations by training a PPO-agent using Python and TensorFlow.

## TECHNICAL SKILLS

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**Languages:** Python, C, C#, C++, SQL, Bash, TypeScript

**Technologies:** Linux, GitHub, Docker, AWS, CI/CD, PyTorch, TensorFlow

## INTERESTS

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- Snowboarding, Competitive Gaming